

Finite element analysis of wideband nanostructures for photovoltaic applications

Niccolò Breda, Giuseppe Pelosi, Stefano Selleri, and Ruggero Taddei

Department of Electronics and Telecommunications, University of Florence,
Via di S. Marta 3, 50139, Florence, Italy

11th International Workshop on Finite Elements for Microwave Engineering – FEM2012
June 4-6, 2012, Estes Park, Colorado, USA

- Objectives:
 - Increase the efficiency of a solar cell through nanostructures, capable of triggering plasmonic resonances
 - Extend the resonant bandwidth of such nanostructures

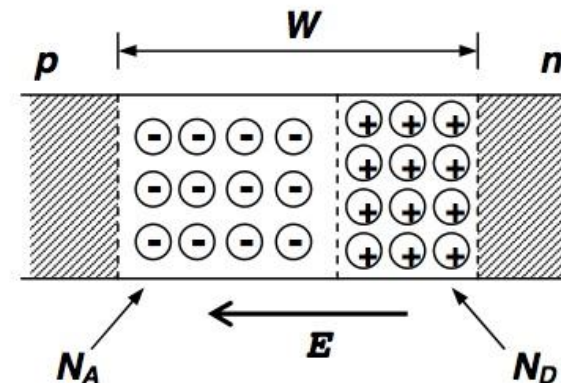
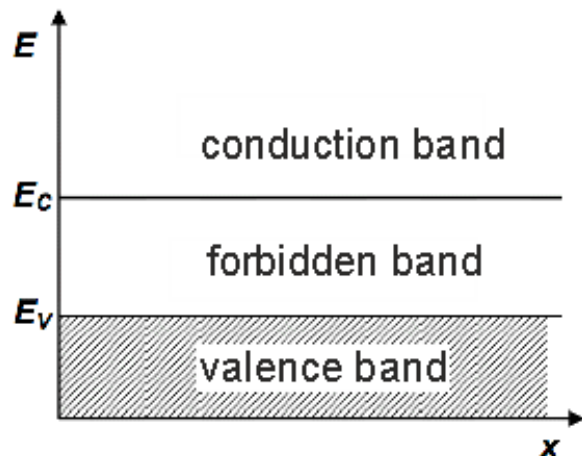
When light is striking over a semiconductor layer, a certain number of electrons passes from the valence band to the conduction band; at the same time an equal number of holes cover the opposite path, reaching the valence band (photoelectric effect).

Only the impinging photons with an energy greater than the bandgap $E = h\nu > E_G = E_C - E_V$ are capable of triggering interband transitions, hence actively foraging the energy conversion process

In the presence of an electric field within the layer, free carriers generated with this process can be exploited to produce a current density.

Electric field in a p-n junction is generated via spontaneous electron-hole recombinations

As a result, a depletion region is formed. A built-in electric field is associated with this region

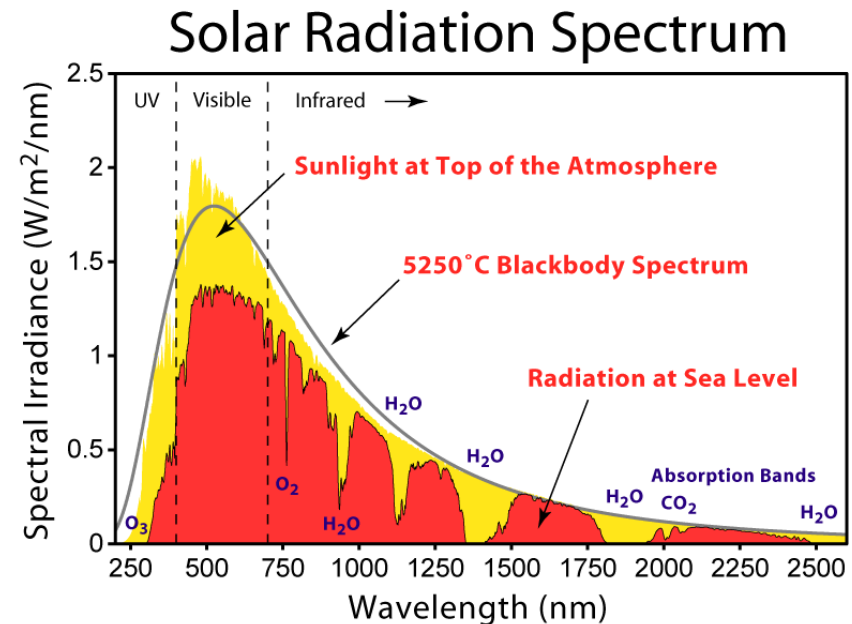


$$\eta = \frac{P_m}{P_{in}}$$

Quantifies the amount of power converted by a solar cell when it is connected to an electric circuit, and in maximum power transfer conditions.

P_m is the power transferred to the load

P_{in} is the impinging spectral power



First and Second Generation Solar Cells

Conventional p-n junctions (1st gen)

Thin film solar cells (2nd gen)

Theoretical limit on efficiency at 33.7 %,
due to the difference in energy between the
bandgap of the substrate (usually silicon) and
solar photons

High manufacturing cost for 1st gen

Lower cost and flexibility for 2nd gen, but
associated with lower efficiency

Third Generation Solar Cells

No need for a silicon-based substrate, hence
no theoretical limit on efficiency

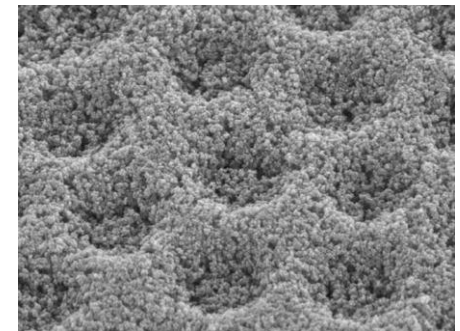
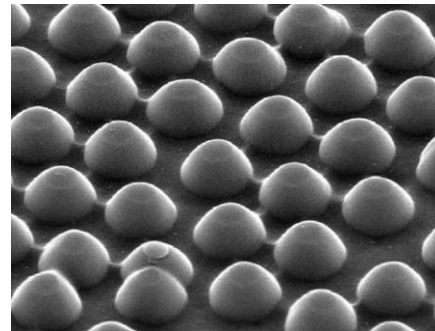
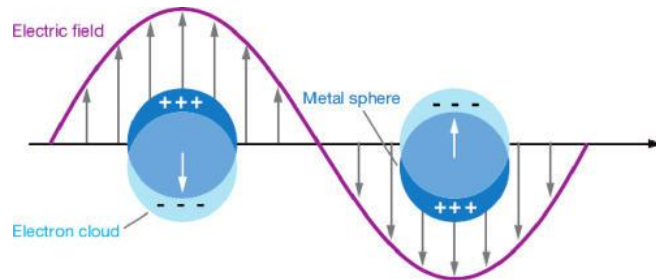
Many possibilities currently under
investigation, including polymer solar cells,
nanocrystalline cells, and dye-sensitized solar
cells

Objectives: low-cost, high efficiency

An innovative approach in the pursuit of an increment in the total conversion efficiency of conventional semiconductor solar cells is emerging. It consists in the exploitation of a deposition of metal nanoparticles over the semiconductor layer, capable of triggering a localized surface plasmon resonance.

Localized surface plasmons (LSP):

Collective oscillations of the electrons in a metal, associated with a great increment of the electromagnetic field in the vicinity of the particle



- **Microwaves:**

metals are reflective: propagation through them is not possible. The perfect electric conductor (PEC) approximation is valid.

- **Near Infrared and Visible:**

penetration of the field into the metal increases, despite the high levels of dispersion

- **Ultraviolet:**

metals acquire dielectric behavior and propagation through them is possible.

Noble metals like gold and silver have their plasma frequency in this range, hence they exhibit a resonant behavior

- Electromagnetic properties of metals have been expressed by the German physicist P.K.L. Drude with a complex dielectric function, derived from a modified plasma model of the metal; at optical frequencies the metal is modeled as a fixed lattice of positive charges, surrounded by a gas of free electrons:

$$\varepsilon_{r,D}(\omega) = \varepsilon_{r,\infty} - \frac{\omega_p^2}{\omega^2 + j\omega\Gamma}$$

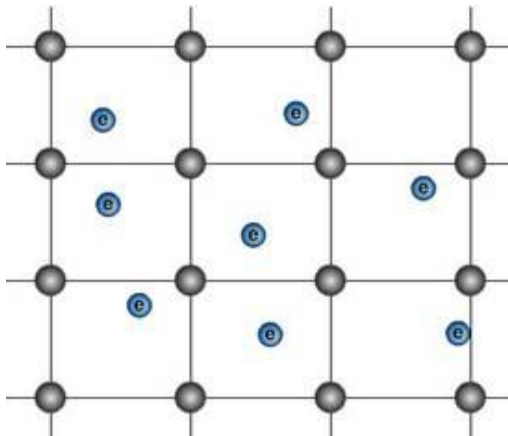
where $\varepsilon_{r,\infty}$ is the complex permittivity at infinite frequencies,

$$\Gamma = \frac{1}{\tau}$$

is the damping rate of the material, defined as the reciprocal of the relaxation time of the electron gas τ , defined as the mean free time between two electron-lattice collisions and

$$\omega_p = \sqrt{\frac{Ne}{\varepsilon_0 m_e}}$$

Is the plasma frequency of the electron gas, N is the electron density, e is the electron charge, m_e is the electron mass



- Making explicit the real and imaginary of the complex permittivity:

$$\varepsilon_r(\omega) = \varepsilon_{r1}(\omega) + j\varepsilon_{r2}(\omega)$$

$$\Re\{\varepsilon_r(\omega)\} = \varepsilon_{r1}(\omega) = 1 - \frac{\omega_p^2 \tau}{1 + \omega^2 \tau^2}$$

$$\Im\{\varepsilon_r(\omega)\} = \varepsilon_{r2}(\omega) = \frac{\omega_p^2 \tau^2}{\omega(1 + \omega^2 \tau^2)}$$

- $\omega \cong \omega_p \Rightarrow \varepsilon_{r1}(\omega) \gg \varepsilon_{r2}(\omega)$

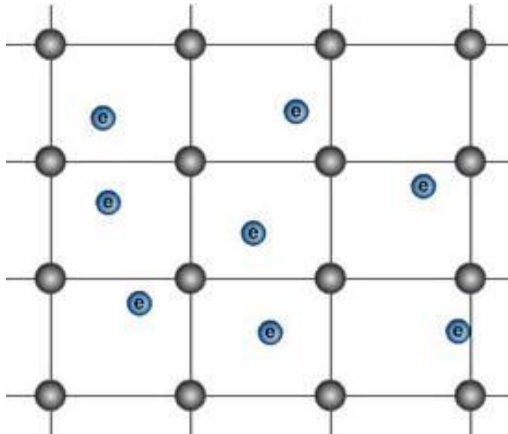
The dielectric function is mostly real, hence propagation is possible through the metal

- $\omega < \omega_p \Rightarrow \varepsilon_{r1}(\omega) \ll \varepsilon_{r2}(\omega)$

The dielectric function is mostly imaginary, hence a damping mechanism occurs: no propagation through the metal at frequencies below the plasma frequency

The simple Drude model does not take into account the interband transitions which occur in metals at frequencies above the plasma one.

The Drude-Lorentz model, which consists in the addition of K Lorentzian terms, corresponding to K damped harmonic oscillators used to describe the small resonances observed in the metal's frequency response, is instead capable of achieving a more accurate modeling of the permittivity at optical frequencies

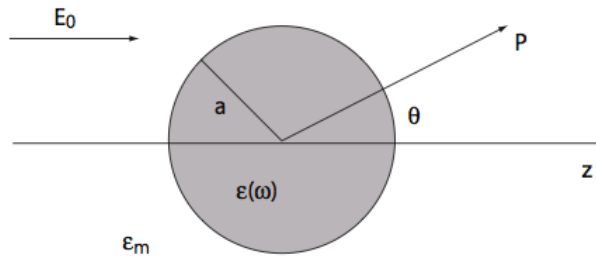


$$\varepsilon_{r,DL}(\omega) = \varepsilon_{r,D}(\omega) + \sum_{j=1}^K \frac{f_j \omega_p^2}{(\omega_j^2 - \omega^2) - j\omega\Gamma_j}$$

where ω_j , f_j , Γ_j are the resonance frequency, strength and damping frequency, respectively, of the k-th oscillator,

and the simple Drude model can be viewed as

$$\varepsilon_{r,D}(\omega) = \varepsilon_{r,\infty} - \frac{f_0 \omega_p^2}{\omega^2 + j\omega\Gamma_0}$$



Localized Surface Plasmons are nonpropagating excitations of the conduction electrons of metallic nanostructures coupled to the electromagnetic field, which arise naturally from the scattering problem of small, subwavelength conductive nanoparticles in an oscillating electromagnetic field

If $a \ll l$, there is no phase variation of the field within the sphere.

Hence an electrostatic approximation is possible. Under these circumstances, the nanoparticle acts as an electric dipole, resonantly absorbing and scattering electromagnetic fields

The plasmonic resonance occurs when the Frölich condition is met $\Re\{\varepsilon(\omega)\} = -2\varepsilon_m$

Which corresponds to a great value of the polarizability

$$\alpha = 4\pi a^3 \frac{\varepsilon - \varepsilon_m}{\varepsilon + 2\varepsilon_m}$$

And hence to a great value of the magnitude of the polarization intensity

$$\mathbf{p} = \varepsilon_0 \varepsilon_m \alpha \mathbf{E}_0$$

At the plasmonic resonance the electric field inside and outside the nanoparticle is greatly enhanced, and so does the efficiency with which a metal nanoparticle scatters and absorbs light

Internal electric field

$$\mathbf{E}_{in} = \frac{3\varepsilon_m}{\varepsilon + 2\varepsilon_m} \mathbf{E}_0$$

External electric field

$$\mathbf{E}_{out} = \mathbf{E}_0 + \frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{p}) - \mathbf{p}}{4\pi\varepsilon_0\varepsilon_m} \frac{1}{r^3}$$

Scattering and Absorption cross-section:

$$C_{sca} = \frac{k^4}{6\pi} |\alpha|^2 \propto a^6$$

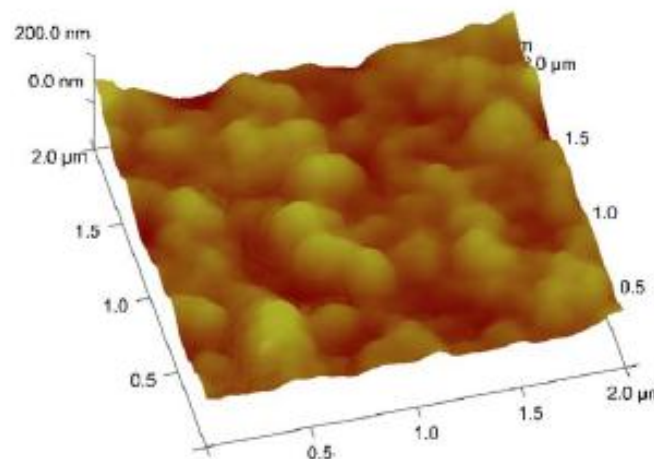
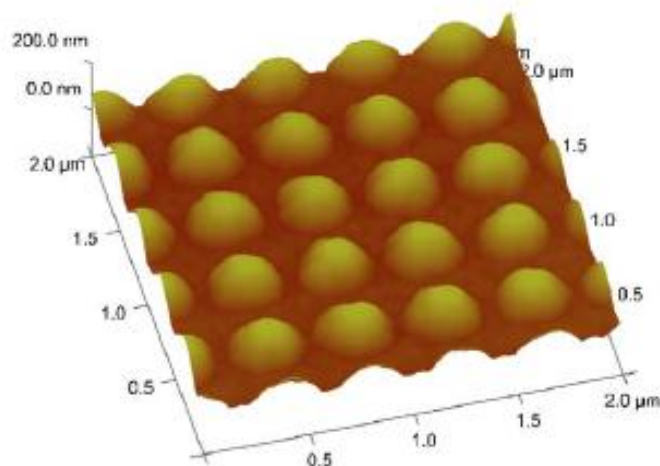
$$C_{abs} = k \operatorname{Im}\{\alpha\} \propto a^3 \quad \text{dominant for small particles} \quad a \ll \lambda$$

Ten to one-hundred-fold increments of the electric field are reported in literature

The properties of a structure composed by a deposition of metallic nanoparticles over a silicon substrate

Regular (periodic array) deposition opposed to random deposition

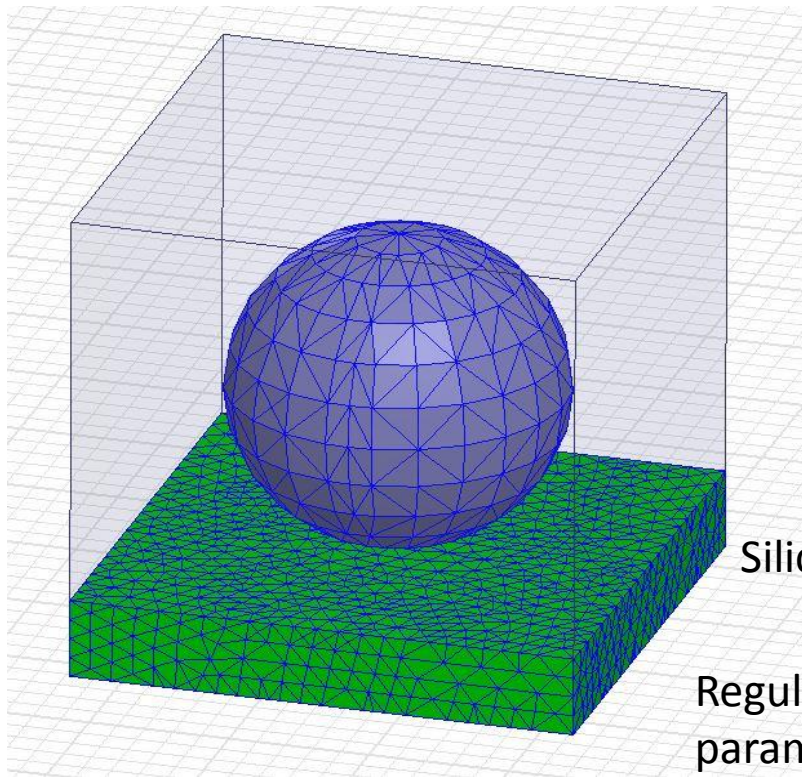
Investigation inspired by Polman's work, where a solar cell composed by a periodic deposition of nanoparticles is compared to a randomly patterned one



Finite Element Setup – Unit Cell

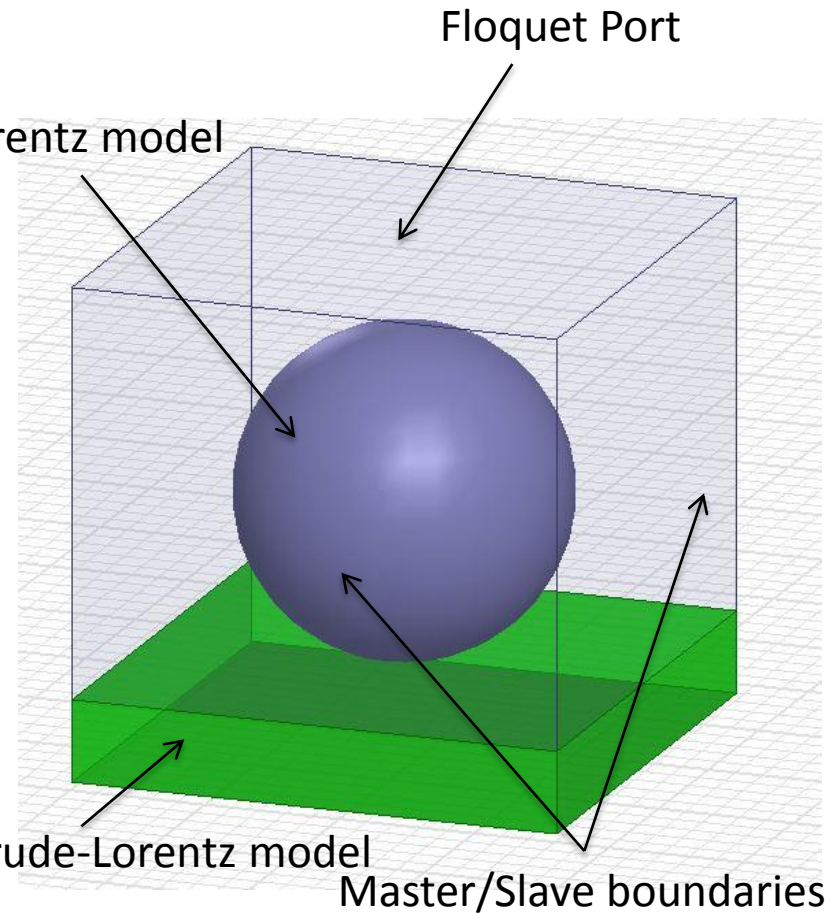
50000+ tetrahedra

First Order Basis Functions

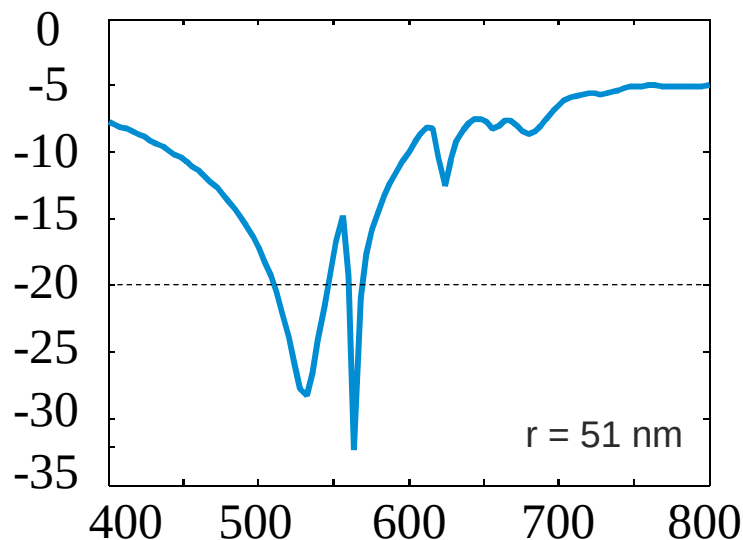


Silver– Drude-Lorentz model

Silicon – Drude-Lorentz model



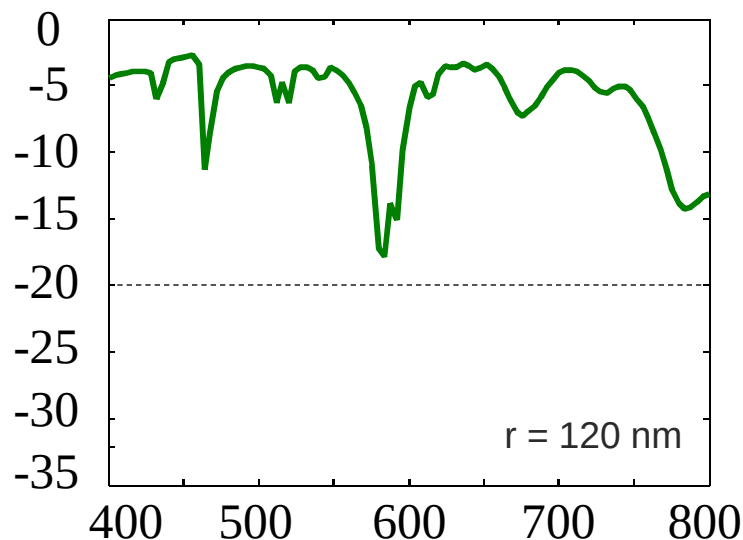
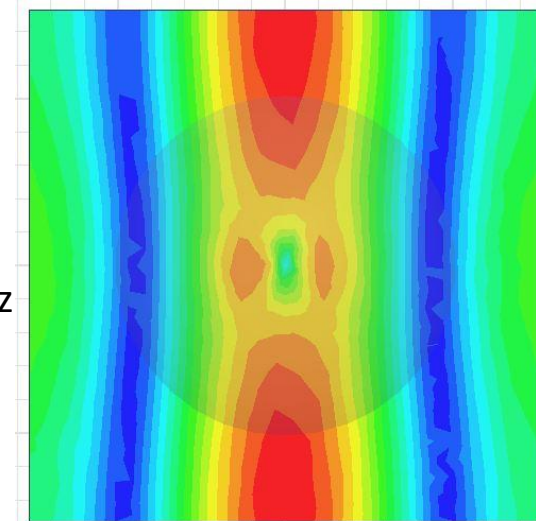
Regular periodic deposition – particle radii chosen after parametric analysis in the [30 – 150] nm range



Radius: 51nm

Mean absorption : - 9.47 dB

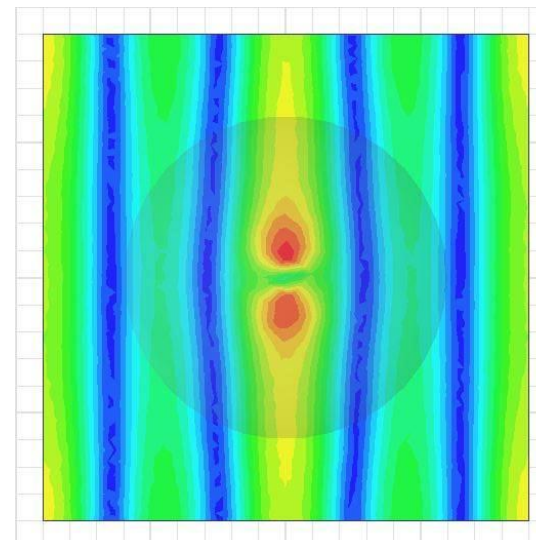
Field enhancement @ 564 THz



Radius: 120nm

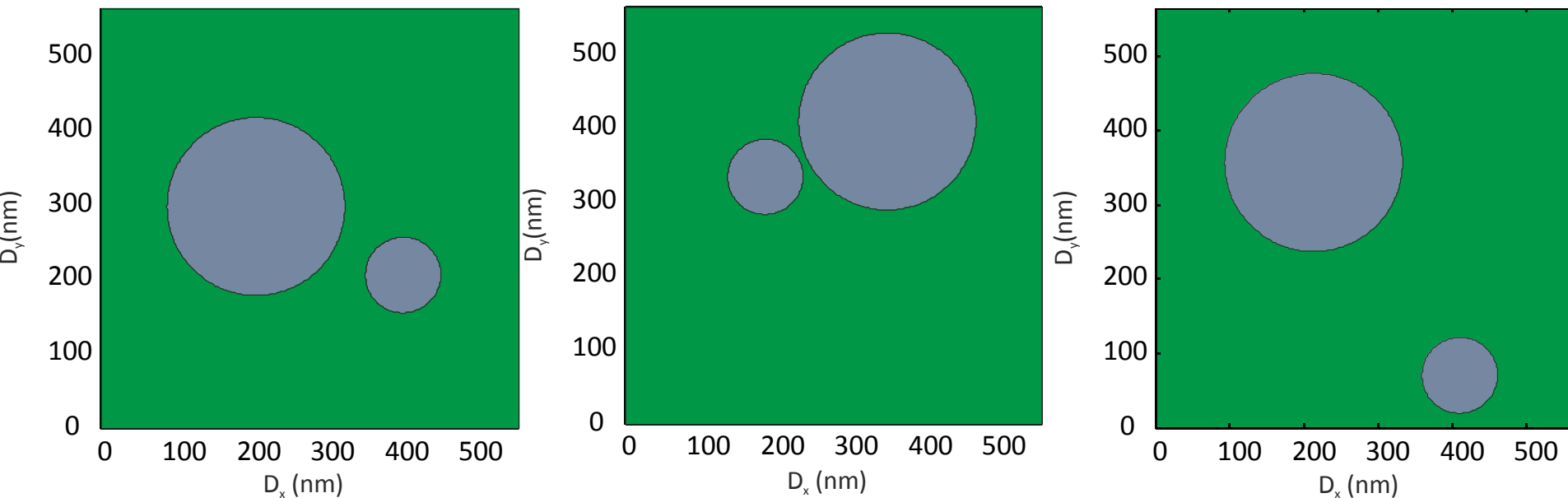
Mean absorption : -5.53 dB

Field enhancement @ 584 THz



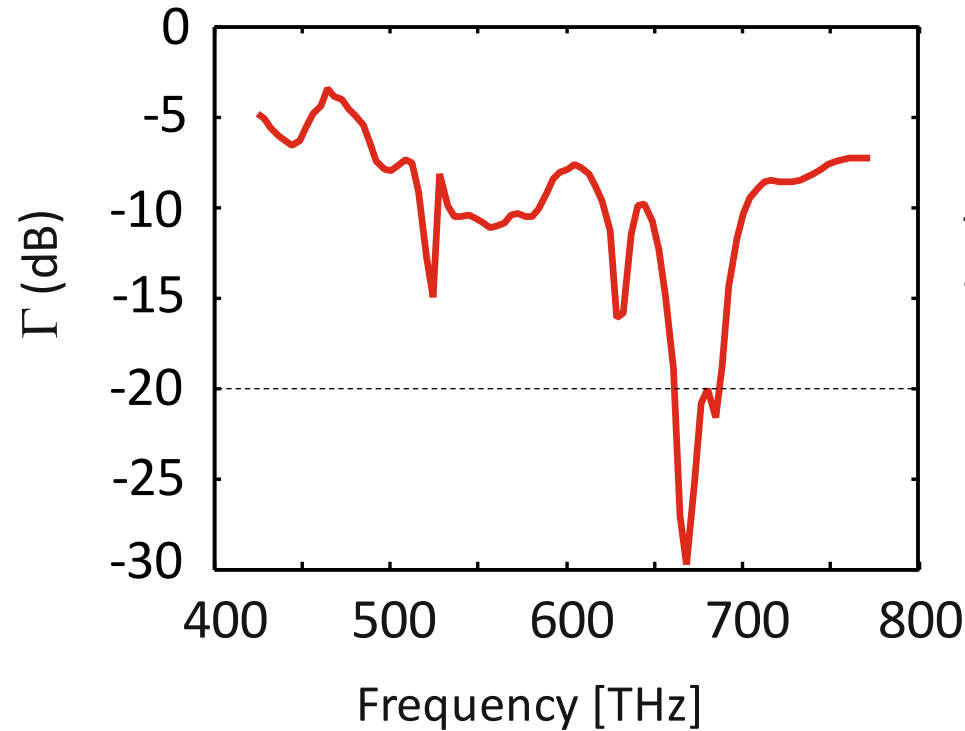
Using particle diameters previously selected, a great number of two-particles deposition configurations has been simulated.

Particle depositions have been chosen in a random fashion, with the same FE setup as before, only with a fixed-dimensions unit cell.

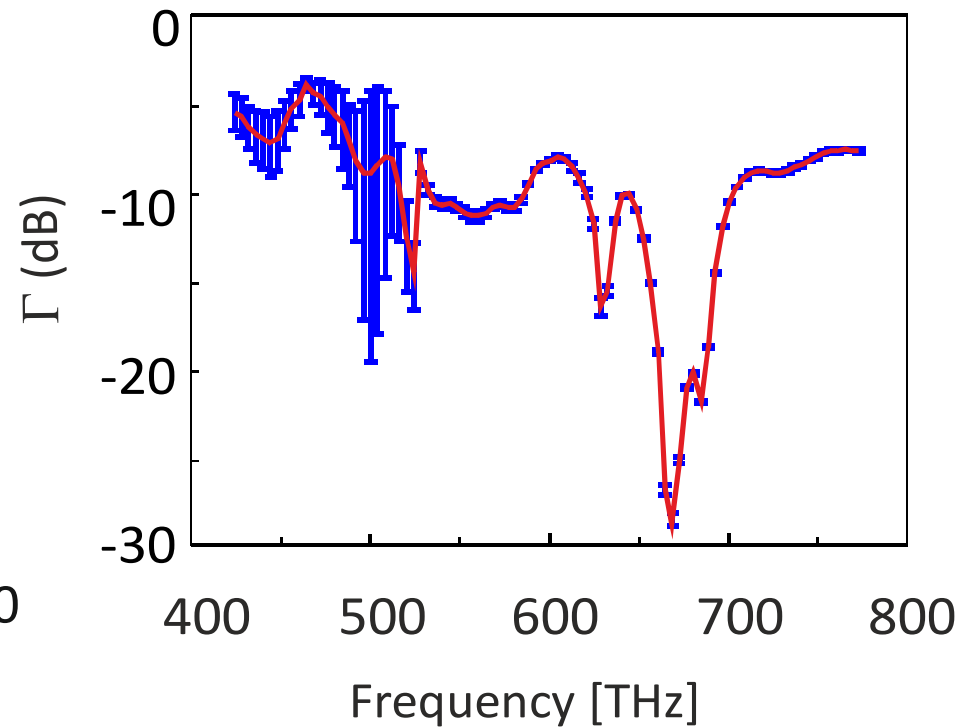


Results

Mean absorption



Error bar



Mean absorption over the frequency range: -9.28 dB

Comparable absorption with the periodic array case

- A FEM simulation with complete Drude-Lorentz model has been carried out
- Parametric analysis over particles of different radii has been performed
- Analyses with particles of different radii with a periodic pseudo-random pattern has been carried out and statistics generated.
- Further investigations especially on particle randomness is necessary